# Applied Linear Algebra

# Least Squares

Devi Prasad M
Manipal School of Information Sciences
Manipal

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# Least Squares

For most choices of $\mathbf{b}$, however, there is no $n$-vector $\mathbf{x}$ for which $\mathbf{A}\mathbf{x} = \mathbf{b}$. As a compromise, we seek an $x$ for which $\mathbf{r} = \mathbf{A}\mathbf{x} - \mathbf{b}$, which we call the *residual* (for the equations $\mathbf{A}\mathbf{x} - \mathbf{b}$), is as small as possible.

We should choose $\mathbf{x}$ so as to minimize the norm of the residual, $\mathbf{A}\mathbf{x} - \mathbf{b}$.

# Least Squares



Minimizing the norm of the residual and its square are the same, so we can just as well minimize

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \|r\|^2 = r_1^2 + \cdots + r_m^2,$$

the sum of squares of the residuals.